

GENERAL SPECIFICATION G3.40

General Scope for Sewer Pipe Laying Works

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G3.40 GENERAL SCOPE FOR SEWER PIPE LAYING WORKS

G3.40.1 Site Conditions

- (a) The Contractor shall be deemed to have inspected the site before submitting his tender. Any claim for extras to or variation from the Contract price resulting from incomplete inspection of the site before submission of tender will not be entertained. The Contractor shall find out any difficulties that may occur, e.g. weather conditions, traffic conditions, difficulty of access, soft ground, obstructions from existing structures, hand rails, trees, drains, or due to proximity of underground utility services, etc and make allowances accordingly in his tender.
- (b) The Contractor shall note that some proposed manholes are located on slopes or inside densely forested area. The Contractor shall take these site conditions into consideration while pricing for laying of sewers and construction of manholes. The Contractor's rates shall be deemed to include all these difficulties and shall not be entertained for any additional cost.
- (c) The Contractor shall ascertain the ground conditions by means of trial holes, bore-logs or otherwise and allow for excavating, working and pipe-laying in adverse conditions such as soft ground, water table, and presence of rock materials.

G3.40.2 Extent of Work

- (a) The Contractor shall ascertain for himself the extent of work that is required to be done on Site and to generally obtain his own information on all matters affecting the execution of the Works including those affecting the type of equipment required to perform the work. The Board will not entertain any claims for extras as a result of ignorance on this respect. The Board assumes no responsibility for any conclusions or interpretations made by the Contractor on the basis of the information made available by the Board.
- (b) The positions of the proposed pipelines / manholes / chambers, etc. shown in the Drawings are only indicative. The Contractor shall excavate trial trenches to establish the most appropriate locations for the proposed pipelines / manholes / chambers, etc. when required and shall submit a setting-out plan indicating the proposed locations / alignments to the S.O. for his approval.
- (c) The Contractor shall perform all necessary site investigation on the location to ascertain for himself the type and depth of any obstruction. The Contractor shall highlight any potential obstructions that may affect the proposed Works and bring this to the S.O.'s attention within the first 6 weeks from the date of site possession.
- (d) The Contractor shall construct the manholes /chambers as shown in the Drawings. The Contractor may propose modification to the manholes / chambers with their PE's endorsement only if Site condition does not permit the manholes /chambers to be constructed. The Contractor is to allow for this item in his rates.
- (e) The Contractor shall be solely responsible for making all the necessary arrangements in coordinating with all the relevant authorities, service departments, specialists, consumers, property owners etc. to ensure satisfactory completion of this Contract.

G3.40.3 Setting Out

- (a) The Contractor shall within two (2) weeks of site possession, engage a registered surveyor to set out the lot boundaries, as indicated in the Drawings. The lot boundaries shall be clearly marked out on the Site using pegs or posts with markings on the ground. The Contractor's Licensed Surveyor and the Contractor shall jointly set out the alignment of the proposed Works.
- (b) The Contractor shall submit a 'setting out' plan duly endorsed by the Surveyor to the S.O. prior to commencement of work. The 'setting out' plan shall show clearly the lot boundaries and alignment and the distance of the proposed manhole/discharge chamber with respect to the lot boundaries.

G3.40.4 Materials to Be Supplied by The Contractor

- (a) The Contractor shall supply all new materials required for successful completion and commission of this Contract.
- (b) All supplied materials shall be as specified, and in compliance with the Singapore Standards, Eurocode Standard Specification, the standards specified herein or any comparable or equal national or international standard acceptable to the S.O. Certificate of Origin for any materials shall be made available on demand.
- (c) Where National Standards are mentioned in these Specifications, the latest editions with up-to-date amendments shall apply, unless otherwise stipulated.
- (d) Where National Standards, Trade names, Proprietary goods or Materials are mentioned in these Specifications, alternative offers of an equal or superior quality of product or workmanship may be employed subject to the prior written approval of the S.O.. Where local products are used, those bearing the SPRING SINGAPORE mark shall be preferred.
- (e) Any material rejected or condemned by the S.O. as unsuitable or not up to standard shall be removed from the site forthwith by the Contractor.
- (f) The Contractor shall arrange for tests of all materials and concrete as specified to be carried out at such laboratories as and when the S.O. may direct. The cost of the samples, tests and transport for the tests shall be borne by the Contractor.

G3.40.5 Evasion

- (a) The whole of the Works shall be carried out and completed in accordance with the true meaning and intent of the Contract, and the Contractor shall not evade or seek extra payment for obvious requirements owing to any discrepancies or omissions in the Drawings, Specifications and Bill of Quantities for the accuracy of which the Contractor shall satisfy himself before sending in his Tender.

G3.40.6 Quantities

- (a) The quantities set out in the Bills of Quantities are ESTIMATED only and shall not be taken as the actual requirement of the S.O. or as limiting or extending the work to be done in accordance with this Contract.
- (b) The Contractor's attention is also drawn to the fact the Bills of Quantities herein are not to be considered sufficiently precise for the ordering of materials. Prior to the placing of any orders for materials for any portion of the Works, the Contractor shall ensure for himself the actual quantity of materials required for the Works.
- (c) All quantities shall be measured throughout the progress of the Works. The S.O. is to be consulted at all times prior to the carrying out of any work or placing of any order for materials.

G3.40.7 Working on Public Roads

- (a) The Contractor shall submit the method of trench excavation for the traffic diversion scheme (if there is any) for all stages of the road works to the S.O. for approval before commencement of work. Traffic diversion schemes shall be prepared in accordance to the latest LTA's Code of Practice for Traffic Control at Work Zone and shall be approved by LTA. A contingency plan encompassing publicity programmes and a work plan to open the affected road with the use of steel decking to relieve traffic build-up whenever necessary must also be in place. All entrances/passages affected by the work shall be made passable to vehicles and / or people at all times.
- (b) Temporary information sign at the worksite, showing type of work, expected completion date, name of Contractor and other details as specified by LTA are to be displayed by the Contractor.
- (c) The Contractor shall note the traffic conditions and pedestrian flow on the Site and shall make his own arrangements, subject to approval of the S.O. and relevant authority, for the diversion of traffic, provision of temporary footpaths, temporary lighting, crossings, protection etc. during the course of his work.
- (d) The Contractor shall take all necessary precautionary measures and provide temporary warning signs and adequate temporary barricades and/or fencing to isolate the worksite from the public. Effort shall be constantly made to ensure the safety of the public in the vicinity of the work. In the event that work is not being carried out, the Contractor shall remove any barricades that are in place and cover the chamber, shaft or trench openings with temporary decking as specified, to allow movement by the public and vehicles. The temporary decking shall be designed to have a minimum skid resistance of 65 British Pendulum Number (BPN).
- (e) The Contractor is to ensure that works to be carried out in these chambers, shafts or trench openings are completed in the shortest possible time and reinstated immediately on completion to minimise disruption to traffic flow and inconvenience to residents and public in the area. No closure of roads will be allowed for the laying of sewer.

- (f) It may be necessary by reason of the safety of the Works, the restoration of interrupted services, or as required by the relevant agencies/authorities or for any other reason whatever, that the Works will need to be carried out continuously by day or by night. Upon notification by the S.O. the Contractor shall carry out such Works, or any required portion thereof, continuously by day and by night. No additional claims by the Contractor will be entertained for this work.
- (g) As required by LTA, the stretch of road affected by the Works shall be milled and patched with minimum thickness of 45mm of W3B mix or 75mm of steel slag mix to the full width of the affected lane to LTA's requirements upon the completion of the pipe laying works.
- (h) The Contractor shall note that construction works affecting road carriageway shall be carried out only during night time and off peak hours all according to LTA requirements. No work shall be carried out during the day time unless otherwise approved by the LTA. Should the work encroach into the road, the Contractor shall provide temporary traffic diversion if required including Traffic Impact Assessment and traffic decking for LTA's approval and such cost shall be deemed to be included in his rates. If decking is required, the Contractor shall provide steel decking and support system for smooth flow of traffic and they shall be designed, endorsed and supervised by the Contractor's Professional Engineer (Civil & Geotechnical) in compliance to LTA and Building Regulations and approved by the LTA, BCA and S.O.. No additional claims by the Contractor will be entertained for this work. The Contractor's Professional Engineer shall apply to BCA and obtain a Temporary Occupation Permit (TOP) before the decking can be opened for use.
- (i) The Contractor shall note that all night work is to be complied with NEA Noise Requirement and other Authority Requirements.
- (j) The Contractor is required to have a valid road-opening permit, which is approved by the LTA. The road-opening permit shall be valid during the period of the Works and the defect liability period of this contract.
- (k) The Contractor shall prepare and submit for the Road Opening Application (ROA). All cost in relation to the ROA, including application fee and daily charge for road works with lane reduction, shall be borne by the Contractor and deemed included in the Contract Sum. The traffic control plan and the relevant supporting document shall be reviewed and endorsed by the appointed civil engineer who possesses the certificate for Basic Traffic Management Course before submitting to the S.O. for onwards submission to LTA.
- (l) The Contractor shall engage a full-time site supervisor on site to supervise road openings, signing arrangement, reinstatement works and to receive directions from LTA. The supervisor shall possess a certificate in Pavement Construction and Maintenance Course issued by the Building Construction Authority.

G3.40.8 Scheduling of Pipelaying Work

- (a) Trench excavation shall not proceed too far ahead of pipe laying especially in metalled road where the trench shall not advance more than two pipe lengths (maximum 10m) ahead of the pipe laying.

- (b) Backfilling and reinstatement works shall not be delayed unnecessarily, and shall not lag behind the pipe laying by more than 30 meters. Road reinstatement shall not lag behind the backfilled trench by more than 50 meters.
- (c) For road crossings each length or cut length of pipe must be laid, joint, wrapped and backfilled before proceeding with the excavation for the next section of pipeline for the crossing. The cycle of excavation, pipe laying, jointing, wrapping and backfilling for each section of pipeline must be carried out non-stop to completion, until the full crossing has been successfully completed.
- (d) For pipe laying along a busy street especially in the City Area, excavation by machine and laying of pipes or specials by mobile crane shall not be carried out during peak traffic hours but preferably at night and during weekends. Pipe laying shall be carried out in sections not exceeding three pipe lengths at a time. Each section shall be laid, welded, wrapped and backfilled before proceeding with the next section.

G3.40.9 Working in Close Proximity to Existing Services

- (a) The Contractor shall ascertain for himself the exact positions of the existing services and ensure that the services are protected against damages during the course of the work. The Contractor shall be responsible for making good all damages to the services as a result of his work, to the satisfaction of the S.O..

G3.40.10 Working in Close Proximity to Water Mains

- (a) If water mains are expected to be encountered, the Contractor shall ensure that trial trenches are carried out to determine the exact position, alignment and levels of the existing trunk water mains before commencement of any works.
- (b) The water mains shall not be subjected to more than 15mm/s peak particle velocity for any works to be carried out in the vicinity of the water mains and appurtenances.
- (c) All necessary precautions and protection measures shall be taken to safeguard and to avoid damage to the water mains during construction works. The costs of repairs to any water main as a result of works carried out by the Contractor shall be borne by the Contractor. The Contractor is also required to indemnify the Board against all losses and claims arising from damage water mains. The Contractor will also be billed for the estimated quantity of water lost from the damaged water mains.
- (d) The Contractor shall be held responsible for any damage or injury caused to any person, property, roads etc. as a result of any bursting of a main caused by their construction works.
- (e) The construction works or activities shall not cover any of the water main manholes or valve chambers and appurtenances.
- (f) The Contractor shall ensure that pipeline is accessible to the Board's staff for maintenance and repair works at all times and there is no obstruction posed by the construction works.

G3.40.11 Working in Close Proximity to Existing Drainage

- (a) The Contractor shall maintain the full discharge capacity of the existing box culverts and drains at all times during the work.

- (b) For drains, culverts affecting the Works, the Contractor shall design, propose, co-ordinate, seek approval and comply with all requirements from Catchment and Waterways Department and the S.O. at the Contractor's own costs. The design inclusive of all drawings and calculations submitted by the Contractor shall be endorsed by a Professional Engineer.
- (c) The Contractor shall exercise due diligence during construction of the Works to ensure the water quality in the water course will not be potentially affected by leakage of chemical grout or slurry from the Works. The Contractor shall continuously monitor for any changes in the water quality in the water course during construction of the Works.

G3.40.12 Working in Close Proximity to Existing Sewers and Other Pipelines

- (a) The Contractor shall note that there are existing sewers, pumping mains, and other pipelines that may affect the proposed sewers and manhole shafts. In particular, existing sewer pipes may require temporary support or diversion at the shaft locations and shall be at the Contractor's own cost.
- (b) The Contractor shall be familiar with PUB (WRN) requirements for pre-condition CCTV sewer inspection survey before work commencement and post-condition CCTV of existing sewers pipes after its completion. The Contractor shall co-ordinate, seek approval and comply with all requirements from PUB (WRN) and the S.O. at the Contractor's own costs.
- (c) All necessary precautions and protection measures shall be taken to safeguard and to avoid damage to the sewers, pumping mains and effluent pipes during construction works. The costs of temporary support, repairs and/or diversion to any sewers, pumping mains and effluent pipes as a result of works carried out by the Contractor shall be borne by the Contractor. The Contractor is also required to indemnify the Board against all losses and claims arising from damage sewers. The Contractor shall allow for the above in his rates including over-pumping as required.
- (d) The Contractor shall design, co-ordinate, seek approval and comply with all requirements from PUB (WRN) and the S.O. at the Contractor's own costs. The design inclusive of all drawings, calculations and submission submitted by the Contractor shall be endorsed by a Professional Engineer (Geotechnical, if applicable).
- (e) The Contractor shall be held responsible for any damage or injury caused to any person, property, roads etc. as a result of damages to the sewers, pumping mains and effluent pipe caused by their construction works.
- (f) The construction works or activities shall not cover any of the sewer manholes or valve chambers and appurtenances.

G3.40.13 Authorities Requirements

- (a) The Contractor shall allow for and abide by all the authorities requirements set out in all Singapore Law, Standard or regulations. This does not preclude compliance with any and all other Authority requirements that may be imposed by these authorities as a pre-requisite to approval for commencement or hand over of the completed Works to these authorities.

- (b) Contractors shall take all necessary precautions according to the requirements of LTA, NParks, PUB, NEA, SLA, JTC, MOM, HDB, PowerGrid, SingTel, StarHub and other relevant authorities/service providers when working near the cables, roads, drains, footpaths, car parks and other similar utilities and/or services or property of the relevant authorities / agencies.

G3.40.14 Sites for Plants, Equipment and Materials

- (a) The Contractor shall note that there will be no special site provided for his plants, equipment and materials. If such sites are required, he shall be entirely responsible for making the necessary arrangements for security fencing, protection, maintenance, access, etc. subject to the approval of the S.O.; Temporary Occupation Licences, rental charges and all costs arising thereof shall be borne by the Contractor. He shall reinstate the lands affected to the satisfaction of the S.O. and land owners on completion of the Works.
- (b) Tree Area Aeration Catch Pits
 - (i) Subsoil catch pits and drains have been or are being constructed in the vicinity of roadside trees (includes trees in car parks) to provide aeration for the trees. These catch pits/ drains may be encountered along the route of the pipeline. The Contractor shall take necessary precaution when working in the vicinity of these aeration catch pits/ drains and shall make good any damage caused.
- (c) Fencing Cut Through
 - (i) Fences crossed by the Works and forming boundaries of plots outside the area occupied by the Works shall not be cut through or destroyed for more than the distance necessary to permit the erection of the new fencing etc. and the Contractor shall make the ends of the cut fences reasonably secure.
- (d) Protection of Adjacent Grounds and Excavation
 - (i) Soil investigation has been carried out for this Contract and borelogs information provided for the Contractor. However, it shall be the responsibility of the Contractor to satisfy himself as to the soil conditions along the route of the pipelines and if he judges it to be necessary, he shall carry out additional soil investigation and location, methodology and number of bore holes are agreed and approved by the S.O..
 - (ii) In general, the Contractor shall assume the soil to be of an unstable nature and the ground water table to be usually high. It shall be the responsibility of the Contractor to maintain stable soil conditions on the grounds adjacent to the excavation. Prior to carrying out any excavation work, the Contractor must design and submit to the S.O. his proposal for the protection of the excavation such as the use of steel sheet piles, soil stabilisation or other methods as necessary. The Contractor is to note that any methods, which require dewatering of ground water will not be accepted.
- (e) Contractor to Clear up Site
 - (i) On completion of Works, the Contractor shall remove all Temporary Works, surplus materials, plant and equipment and shall leave the Site in a clean and tidy condition to the entire satisfaction of the S.O..

- (ii) All reinstatement of the Site shall be carried out by the Contractor to the entire satisfaction of the S.O.. The reinstated Site shall continue to be maintained by the Contractor for a period as specified in the Contract or as required by the relevant authorities.
 - (iii) The taking over of the Works by the S.O. shall not relieve the Contractor of his responsibility for maintenance after the date of taking over certificate for the full Period of Maintenance stipulated in the Conditions of Contract.
 - (iv) All rubbish shall be cleared from the site and buildings under construction from time to time. The Contractor shall provide purpose made bulk bins at locations approved by the S.O. and shall deposit the waste and debris into these bins and shall make necessary arrangements for ultimate disposal of these wastes and debris at approved dumping grounds. The bins shall be designed to carry debris of density 1.6 T/cu m and preferably be of volume 3 cu m.
- (f) Services to be Provided
 - (i) The Contractor shall make his own arrangements for the supply of electricity, potable water, telephone, compressed air, toilet and other services to the Works, Temporary Works and Construction Plant. Under no circumstances will the Contractor be permitted to use power supply, potable water, telephone, compressed air, toilet and other services supplied to the existing plant and equipment unless otherwise approved by the S.O.. The Contractor shall provide and maintain all pipes, cables and fittings etc. that may be necessary to carry such services to his operations. The whole costs of installations, use, charges and their maintenance etc. during the period of the Contract and the removal on completion of the Contract shall be deemed included in his tender prices.
- (g) Provision of Temporary Lighting
 - (i) Where natural lighting is inadequate, artificial lighting shall be provided at all workplaces and their approaches, including passageways. The general work areas shall be lighted to not less than 100 lux while any work is in progress. The passageways shall be lighted to at least 50 lux.
 - (ii) Workers shall not enter unlit work places without a portable light.
 - (iii) Artificial lighting shall be installed or positioned to avoid glare (both direct and reflected) and prevent shadows from being cast on the work area. Localised lighting shall be positioned at a sufficient height above the working area so as to provide uniform lighting. Lamps shall be protected by suitable guards where necessary to prevent breakages.
- (h) Digitised As-Constructed Drawings
 - (i) As-constructed drawings shall be endorsed "As-Constructed" by the respective Professional Engineer/Registered Surveyor on one set of original plotted drawings and submitted together with their soft copy in CD-R media. All as-constructed drawings both hard and soft copy shall be submitted not later than four weeks after the date of completion of the Works. The Contractor shall endorse and submit the checklist as shown in Appendix V.

- (ii) The as-constructed site plan drawings shall follow the SVY21 coordinated cadastral base established by the SLA. The Contractor shall indicate all horizontal control points (Control Survey Marker) obtained from SLA in the drawings. The co-ordinates of a point are given with reference to the WGS 84 spheroid (Latitude and Longitude) and the grid co-ordinates on the SVY21 datum (Northing and Easting).
- (iii) The Contractor shall submit soft copy of the as-constructed drawings in graphics and scanned formats. The graphics format shall be in "dgn" format or other approved compatible graphics format suitable for use on the CAD software i.e. Microstation or its latest release. The scanned format shall be in tiff or pdf formats and based on the original endorsed as-constructed drawings.
- (iv) The as-constructed graphics format shall follow the 'levels, weight, colour, line style, sheet numbering etc. of the original data file supplied to the Contractor. All information in the as-constructed graphics format shall be in one file without any x-reference. The Contractor shall also submit a soft copy in text files of the as-constructed details using the template shown in appendix VII.
- (v) The Contractor shall also submit a soft copy of the drawing list using the template shown in the Appendix VII.
- (vi) To assist the successful Contractor in expediting the production of shop drawings, as-constructed drawings and any other drawings as required by the S.O., the Contractor may obtain free of charge a soft copy of the tender drawings upon award of the contract. The Contractor shall supply, new CD-R/DVD-R for this purpose. The CD-R/DVD-R supplied shall be sufficient to store all the required files.
- (vii) The soft copies of the as-constructed drawings will only be accepted if a test plot of each of the drawings on the CAD System is successful. Upon acceptance, the drawings and its storage media shall become the sole property of the Board. All related cost associated with complying with these requirements is deemed to have been allowed for in the tender document, whether indicated explicitly or otherwise.
- (viii) It shall be clearly understood, by the Contractor, that the provisions spelt out under the clauses with the heading of 'DIGITISED AS-CONSTRUCTED DRAWINGS' (herewith referred to as "THESE CLAUSES"), does not absolve the Contractor of his responsibility for the production and timely submission of the as-constructed drawings or any other drawings requested by the S.O. and also as required under the terms of this contract. The provisions under THESE CLAUSES are merely to facilitate the production of these drawings.
- (ix) In the event that the Contractor intends to obtain the soft copy of the tender drawings, the Contractor is required to put forth his/her request in writing to the S.O. within three months from the date of site possession.

- (x) The Contractor shall note that the provisions of THESE CLAUSES for soft copy of the tender drawing files are not an entitlement and it's provisions are meant solely for the expeditious completion of the requirements of this contract. It shall not be used for and in any other purpose without the written permission of the Board. The Contractor shall not, in anyway whatsoever, re-use them or sell, hire and lend these files to another party or person without the Board's prior written approval. The Board shall not bear any responsibility for the accuracy and omissions found in the contents of these files and also that the Board shall be indemnified against any claims, damages, expenses and cost (including third parties) that may arise, whether directly or indirectly, from the use of the contents of these files. It shall also be deemed that the Contractor have read and understood the disclaimer's contents of THESE CLAUSES upon acceptance of the work as spelt out under this contract.
- (xi) An item has been allowed for in the Bills of Quantities for the Contractor to price for complying with the above-mentioned requirements.
- (i) Guarantees from manufacturers
 - (i) The Contractor shall obtain from such manufacturers and specialists as directed by the S.O. guarantees covering the workmanship and materials for the periods agreed. The original guarantee must be forwarded to the S.O. two weeks before completion of such works.
- (j) Working Space on State Land
 - (i) The working space mentioned above is subject to availability and the Contractor is to ascertain for himself the availability of such space. Upon award of tender, the Contractor shall submit details of working space for the S.O.'s approval prior to commencement of work. The Contractor shall be required to confine his work within the working space approved by the S.O. or other relevant authorities. The Contractor is to apply for a TOL from the Singapore Land Authority and the Contractor is to bear the TOL fee and other charges imposed by Singapore Land Authority. No claims whatsoever in this respect shall be entertained by the Board. The Contractor is to note that all approved working space shall be used solely for the purpose of construction works under this Contract. No site office, storage of materials and equipment and workers quarters are permitted within the approved working space.
 - (ii) All working space used by the Contractor shall be reinstated to the satisfaction of the S.O. and the other relevant authorities upon completion of the Works.
- (k) Code of Practice for Works
 - (i) The whole of the Works to be carried out under this Contract is to be executed in accordance with the relevant Singapore Standard and Code of Practice & Eurocode Standard code of practice.
 - (ii) The Singapore Standard and Code of Practice are available at SPRING Singapore.

- (l) Reinstatement for the Work Sites
 - (i) Unless stated otherwise, the Contractor shall reinstate all roads and footways affected by the Works to the extent, lines and levels that existed prior to the commencement of the Works and to standards that are at least equivalent to those that existed prior to the commencement of the Works.
 - (ii) Unless stated otherwise, the Contractor shall reinstate all surface water drainage systems (including but not restricted to gullies, channels, catch pits, pipe runs, drop inlet chambers, scupper pipes, manholes and covers and the like) affected by the Works. The standard of reinstatement shall be at least equivalent to that existing prior to the Contract commencement and in accordance to relevant authorities' requirements.
- (m) Complying With Requirements of the Authorities and Agencies
 - (i) The Contractor shall comply with all the requirements of the authorities and agencies such as BCA, LTA, NParks, SLA, PUB, NEA, URA, JTC, HDB and Power Grid, etc. The Board accepts no responsibility for the completeness and accuracy of the information. The Contractor shall obtain from the relevant authorities all the requirements that they have to comply in respect of their services prior to commencement of Works and during the construction of the Works and comply with all such requirements.
 - (ii) The Contractor's attention is drawn to PowerGrid and telecommunication cables. If construction works are affected by PowerGrid's and telecommunication cables, the Contractor shall seek S.O. and PowerGrid or telecommunication providers for advice. In the event where crossing of PowerGrid and telecommunication cables are necessary, the Contractor shall liaise with PowerGrid or telecommunication providers and seek assistance in protecting, suspending the cables, and supervising the works so as to protect against damage.
 - (iii) The Contractor's Professional Engineer (Geotechnical, if applicable) shall design for suspension of cables, Temporary Works, jet grouting to stabilize the gaps, excavations so required for the protection of cables against damage. All major cables shall not be diverted and costs associated with the suspension of the major cables shall deem to be included in the tender.
 - (iv) The Contractor's attention is drawn to the services which may be present on site; such as PowerGrid cables, PUB water mains, Used water pipes, Drainage Crossings and telecommunication cables.
 - (v) The Contractor shall ascertain the locations of such services and include the costs for carrying out the Works in the vicinity of such services.
- (n) Catchment and Waterways Requirements
 - (i) The execution of any work shall not change, disrupt, fill, block, divert or disturb the existing overland flow or the existing system of drains unless an alternative drainage system has been approved by the Board.
 - (ii) The runoff within, upstream of and adjacent to the worksite shall be effectively drained away without causing drainage problems within the worksite or in areas outside the worksite.

- (iii) All earth slopes shall be set outside the Drainage Reserve.
- (iv) No obstruction of any kind shall be placed, laid or erected within Drainage Reserves and drains unless authorised by the Board. If so, the Drainage Reserves and drains shall remain accessible at all times to workers and plant/equipment of, or authorised by the Board.
- (v) No temporary structures such as cofferdams, crossings, stagings, etc. shall be constructed in or over an existing drain without the approval of the Board.
- (vi) Any temporary structure if approved within a drain or Drainage Reserve shall be removed immediately on completion of the construction works or as directed.
- (vii) Drainage inlets such as drop-inlet chambers and scupper drains shall not be blocked by any material or debris.
- (viii) Drains shall not be covered without the approval of the Board.
- (ix) Bunds of stockpiled materials such as earth from trench work shall not be longer than 10 m and gaps of at least 1 m width shall be provided between the bunds to allow the free flow of surface runoff.
- (x) Material from any stockpile shall not be allowed to fall or be washed into the drain. Adequate preventive measures, including the provision of proper and stable barricades or screens where necessary, shall be provided.
- (xi) Any damage to existing drain/drainage facilities arising out of any work shall be immediately reported to the Board. Any obstruction to flow arising from the damage shall be immediately and completely cleared. The damaged drain shall be repaired or reinstated to the requirements of the Board.
- (xii) No services shall be laid or left (in conjunction with development works) within a Drainage Reserve. Where the services have to cross a Drainage Reserve, approval of the Board shall be obtained.
- (xiii) The existing drainage system shall not be altered or interfered with unless specific written approval is obtained from the Board.
- (o) Working Site
 - (i) The Contractor shall confine all his operations, plant, equipment and stores to the approved working site. He shall provide reasonable facilities for access to the working site.
- (p) Fencing and Gates, etc
 - (i) Provide, erect, alter as required and maintain all necessary screens, gates, covered walkways, footways, gantries, temporary enclosures, barriers, etc. for the proper execution of the Works and the protection of the public in accordance with the authorities' regulations and all to the satisfaction of the S.O.. The fencing is to be maintained to the highest standards throughout the duration of the Contract, including providing replacement and extension materials and including anti-terminate treatment as necessary.

(q) Defects Liability Period

- (i) The Defects Liability Period (DLP) shall be commence from the date of substantial completion of the Works as certified by the S.O.. Where works such as reinstatement works are to be handed over to the relevant authorities or other parties, the Defects Liability Period for such works shall be that imposed by the relevant authorities or other parties.
- (ii) The Contractor is required to carry out CCTV inspection of the pipelines and submit CCTV inspection reports in accordance to the specifications, on all completed pipelines prior to expiry of the DLP. An item for this work has been allowed for in the Bills of Quantities. The Contractor shall require to carry out all rectification works such as repairs or replacement of defective works, etc as the Superintending Officer may instruct in writing during this period or within thirty (30) days of the expiry of the Defects Liability Period. The costs of any CCTV re-inspections required to verify that defects found in the completed sewers during DLP that have been rectified shall be borne by the Contractor. Should the Contractor fail to comply within seven (7) days with this written instruction, then the Board reserves the right to engage others to rectify the works without prior notice to the Contractor and all costs and expenses incurred, including 20% administrative charges shall be borne by the Contractor.

G3.40.15 Contract Drawings

- (a) The Specification shall be read in conjunction with the Drawings in Volume 4 of the tender documents.
- (b) Further details and Drawings, in amplification thereof, may be issued by the S.O. from time to time during the progress of the Works.
- (c) Sewerage Standard Drawings, which form part of the tender drawings, are available for downloading at the Board's website, <http://www.pub.gov.sg>

END OF SECTION